

Time Series Forecasting Project Expose

Group ID: 140

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1 Idea

The goal of this project is to investigate deep learning approaches for multivariate time forecasting by predicting future data load and evaluating it on the provided benchmark dataset. In our approach we aim to compare recurrent and attention-based deep learning architectures and measure their feasibility in terms of short-term dynamics and long-range dependency capturing.

2 Planned architectures

We are considering two main model families for implementation and evaluation:

1. LST + attention

The first approach is an encoder-decoder architecture model based on the Long Short- Term Memory networks combined with an attention mechanism. The motivation behind this choice comes as a result of LSTM models capturing sequential temporal dynamics on different states, while attention modules help focus on the most important timesteps of a sequence and generating repeated patterns.

2. Transformer / TCN

3 Additional Dataset

To check if the model generalizes to other data domains beyond the benchmark, we will use a multivariate weather forecasting dataset such as the Jena Climate Dataset that contains measurements across many factors (temperature, humidity, wind speed, etc). The format and the cross-variable dependencies that affect weather make the comparison between LSTM-attention and more realistic and effective regarding future predictions.

4 Planned improvements

Potential improvements and experiments on the two architectures might include: teacher forcing for autoregression, different dropout and regularization strategies, ensemble forecasting, sequence series normalization or learning rate scheduling based on gradient behaviour.

To obtain results, the models will be compared against the provided benchmark baselines mentioned in the source code, including: Naive last-value forecast, Lag-24/-168 repeat and the Seasonal mean baseline. Additionally performance will be measured using the leader board metrics: MAE, RMSE, MAPE, etc.

5 Risks and Open questions

A challenge that might arise during implementation is the risk of error propagation and accumulation across the network as well as the difficulty of forecasting highly noisy data with little correlation. A general problem is related to the trade-off between model complexity and generalization, running to the risk of overfitting due to the dataset size. For all challenges, the main question remains whether transformer or recurrent architectures perform better under the following circumstances.

6 Sources